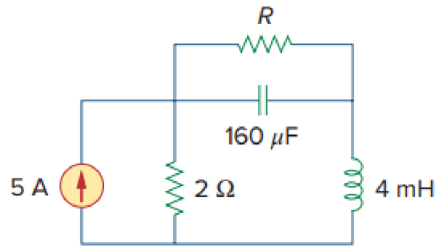
Chapter 6, Problem 47P

Problem

For the circuit in Fig. 6.70, calculate the value of *R* that will make the energy stored in the capacitor the same as that stored in the inductor under dc conditions.

Fig. 6.70:



Step-by-step solution

Step 1 of 5

Consider the following circuit:

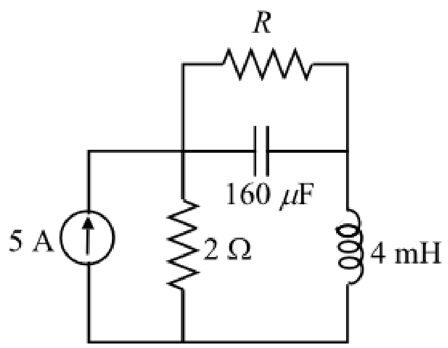


Figure 1

Step 2 of 5

Under dc conditions, the capacitor acts as an open circuit and an inductor acts as a short circuit. Redraw the circuit in Figure 1 under dc conditions.

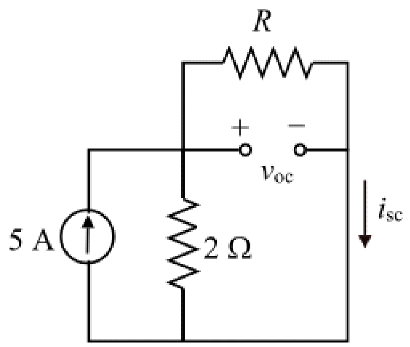


Figure 2

Step 3 of 5

Determine the short circuit current, i_{sc} using current division rule.

$$i_{sc} = (5)\left(\frac{2}{2+R}\right) = \frac{10}{2+R}$$

Determine the open circuit voltage, which is the voltage across the resistance *R*.

$$v_{oc} = i_{sc}R = \frac{10R}{2+R}$$

Step 4 of 5

Determine the energy stored in the inductor under dc conditions.

$$\begin{aligned} w_L &= \frac{1}{2}Li_{sc}^2 \\ &= \frac{1}{2}(4\times10^{-3})\left(\frac{10}{2+R}\right)^2 \\ &= \frac{0.2}{(2+R)^2} \end{aligned}$$

Determine the energy stored in the capacitor under dc conditions.

$$\begin{aligned} w_C &= \frac{1}{2}Cv_{oc}^2 \\ &= \frac{1}{2}(160\times10^{-6})\left(\frac{10R}{2+R}\right)^2 \\ &= \frac{8\times10^{-3}R^2}{(2+R)^2} \end{aligned}$$

Step 5 of 5

To determine the value of *R* at which the energy stored in the inductor is equal to the energy stored in the capacitor under dc conditions, equate both energies.

$$\frac{0.2}{(2+R)^2} = \frac{8\times10^{-3}R^2}{(2+R)^2}$$

$$0.2 = 8\times10^{-3}R^2$$

$$R^2 = \frac{0.2}{8\times10^{-3}}$$

$$R^2 = 25$$

$$R = 5\ \Omega$$

Thus, the value of resistance *R* that makes the energy stored in the capacitor equal to that stored in the inductor under dc conditions is $\boxed{5\ \Omega}$.